## 极小数值型的亏格下界

**Editorial reading note.** Author text is retained verbatim. In the grouping paragraph Phi_j is a source notation slip for Psi_k; the zero-length-chain convention is the preceding parenthetical convention. No new proof is supplied.

**Counting scope.** The primary problem is Tag 0C7C, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

Copyright (C) 2005–2025 Johan de Jong. Stacks Project authors. Source: Tag 0C7C.

Difficulty H2: 本题对极小数值型证明算术亏格不小于关联图的拓扑亏格。逐点势函数非负并不成立，证明须识别负项、沿中性顶点组成极大链并把负贡献分配到终点，再用重数增长和分情形估计排除剩余负项。给定数值型与极小性定义后，该组合估计仍是实质任务，判 H2；不以定义来源或篇幅判 H3。

.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, models.tex, lines 694–843. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

**Definition 0.1.** A *numerical type* $T$ is given by

$$n, m_i, a_{ij}, w_i, g_i$$

where $n \geq 1$ is an integer and $m_i$, $a_{ij}$, $w_i$, $g_i$ are integers for $1 \leq i, j \leq n$ subject to the following conditions

(1) $m_i > 0$, $w_i > 0$, $g_i \geq 0$,
(2) the matrix $A = (a_{ij})$ is symmetric and $a_{ij} \geq 0$ for $i \neq j$,
(3) there is no proper nonempty subset $I \subset \{1, \ldots, n\}$ such that $a_{ij} = 0$ for $i \in I$, $j \notin I$,
(4) for each $i$ we have $\sum_j a_{ij} m_j = 0$, and
(5) $w_i | a_{ij}$.

**Definition 0.2.** We say $n, m_i, a_{ij}, w_i, g_i$ is a *numerical type of genus* $g$ if $g = 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ is the integer from Lemma lemma genus (Tag 0C71).

**Lemma 0.3.** *Let $n, m_i, a_{ij}, w_i, g_i$ be a numerical type. Let $e$ be the number of pairs $(i, j)$ with $i < j$ and $a_{ij} > 0$. Then the expression $g_{top} = 1 - n + e$ is $\geq 0$.*

**Definition 0.4.** Let $n, m_i, a_{ij}, w_i, g_i$ be a numerical type $T$. The *topological genus of* $T$ is the nonnegative integer $g_{top} = 1 - n + e$ from Lemma 0.3.

**Definition 0.5.** We say the numerical type $n, m_i, a_{ij}, w_i, g_i$ of genus $g$ is *minimal* if there does not exist an $i$ with $g_i = 0$ and $a_{ii} = -w_i$, in other words, if there does not exist a $(-1)$-index.

**Lemma 0.6.** *Let $n, m_i, a_{ij}, w_i, g_i$ be a numerical type of genus $g$. If $n > 1$, then $a_{ii} < 0$ for all $i$.*

证明. Lemma lemma recurring symmetric real (Tag 0C5X) applies to the matrix $A$. □

**Lemma 0.7.** *Let $n, m_i, a_{ij}, w_i, g_i$ be a numerical type of genus $g$. Assume $n > 1$. If $i$ is such that the contribution $m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ to the genus $g$ is $< 0$, then $g_i = 0$ and $a_{ii} = -w_i$.*

证明. Follows immediately from Lemma 0.6 and $w_i > 0$, $g_i \geq 0$, and $w_i | a_{ii}$. □

**Lemma 0.8.** *If $n, m_i, a_{ij}, w_i, g_i$ is a minimal numerical type with $n > 1$, then $g \geq 1$.*

证明. This is true because $g = 1 + \sum \Phi_i$ with $\Phi_i = m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ nonnegative by Lemma 0.7 and the definition of minimal types. □

## Primary problem: Tag 0C7C

**Lemma 0.9.** *If $n, m_i, a_{ij}, w_i, g_i$ is a minimal numerical type with $n > 1$, then $g \geq g_{top}$.*

证明. The reader who is only interested in the case of numerical types associated to proper regular models can skip this proof as we will reprove this in the geometric situation later. We can write

$$g_{top} = 1 - n + \frac{1}{2}\sum\nolimits_{a_{ij}>0} 1 = 1 + \sum\nolimits_i (-1 + \frac{1}{2}\sum\nolimits_{j \neq i,\ a_{ij}>0} 1)$$

On the other hand, we have

$$\begin{aligned} g &= 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) \\ &= 1 + \sum m_i w_i g_i - \sum m_i w_i + \frac{1}{2}\sum\nolimits_{i \neq j} a_{ij} m_j \\ &= 1 + \sum\nolimits_i m_i w_i (-1 + g_i + \frac{1}{2}\sum\nolimits_{j \neq i} \frac{a_{ij}}{w_i}) \end{aligned}$$

The first equality is the definition, the second equality uses that $\sum a_{ij} m_j = 0$, and the last equality uses that uses $a_{ij} = a_{ji}$ and switching order of summation. Comparing with the formula for $g_{top}$ we conclude that the lemma holds if

$$\Psi_i = m_i w_i (-1 + g_i + \frac{1}{2}\sum\nolimits_{j \neq i} \frac{a_{ij}}{w_i}) - (-1 + \frac{1}{2}\sum\nolimits_{j \neq i,\ a_{ij}>0} 1)$$

is $\geq 0$ for each $i$. However, this may not be the case. Let us analyze for which indices we can have $\Psi_i < 0$. First, observe that

$$(-1 + g_i + \frac{1}{2}\sum\nolimits_{j \neq i} \frac{a_{ij}}{w_i}) \geq (-1 + \frac{1}{2}\sum\nolimits_{j \neq i,\ a_{ij}>0} 1)$$

because $a_{ij}/w_i$ is a nonnegative integer. Since $m_i w_i$ is a positive integer we conclude that $\Psi_i \geq 0$ as soon as either $m_i w_i = 1$ or the left hand side of the inequality is $\geq 0$ which happens if $g_i > 0$, or $a_{ij} > 0$ for at least two indices $j$, or if there is a $j$ with $a_{ij} > w_i$. Thus

$$P = \{i : \Psi_i < 0\}$$

is the set of indices $i$ such that $m_i w_i > 1$, $g_i = 0$, $a_{ij} > 0$ for a unique $j$, and $a_{ij} = w_i$ for this $j$. Moreover

$$i \in P \Rightarrow \Psi_i = \frac{1}{2}(-m_i w_i + 1)$$

The strategy of proof is to show that given $i \in P$ we can borrow a bit from $\Psi_j$ where $j$ is the neighbour of $i$, i.e., $a_{ij} > 0$. However, this won't quite work because $j$ may be an index with $\Psi_j = 0$.

Consider the set

$$Z = \{j : g_j = 0 \text{ and } j \text{ has exactly two neighbours } i, k \text{ with } a_{ij} = w_j = a_{jk}\}$$

For $j \in Z$ we have $\Psi_j = 0$. We will consider sequences $M = (i, j_1, \ldots, j_s)$ where $s \geq 0$, $i \in P$, $j_1, \ldots, j_s \in Z$, and $a_{ij_1} > 0, a_{j_1 j_2} > 0, \ldots, a_{j_{s-1} j_s} > 0$. If our numerical type consists of two indices

which are in $P$ or more generally if our numerical type consists of two indices which are in $P$ and all other indices in $Z$, then $g_{top} = 0$ and we win by Lemma 0.8. We may and do discard these cases.

Let $M = (i, j_1, \ldots, j_s)$ be a maximal sequence and let $k$ be the second neighbour of $j_s$. (If $s = 0$, then $k$ is the unique neighbour of $i$.) By maximality $k \notin Z$ and by what we just said $k \notin P$. Observe that $w_i = a_{ij_1} = w_{j_1} = a_{j_1j_2} = \ldots = w_{j_s} = a_{j_sk}$. Looking at the definition of a numerical type we see that

$$\begin{aligned} m_i a_{ii} + m_{j_1} w_i &= 0, \\ m_i w_i + m_{j_1} a_{j_1 j_1} + m_{j_2} w_i &= 0, \\ &\ldots\ldots \\ m_{j_{s-1}} w_i + m_{j_s} a_{j_s j_s} + m_k w_i &= 0 \end{aligned}$$

The first equality implies $m_{j_1} \geq 2m_i$ because the numerical type is minimal. Then the second equality implies $m_{j_2} \geq 3m_i$, and so on. In any case, we conclude that $m_k \geq 2m_i$ (including when $s = 0$).

Let $k$ be an index such that we have a $t > 0$ and pairwise distinct maximal sequences $M_1, \ldots, M_t$ as above, with $M_b = (i_b, j_{b,1}, \ldots, j_{b,s_b})$ such that $k$ is a neighbour of $j_{b,s_b}$ for $b = 1, \ldots, t$. We will show that $\Phi_j + \sum_{b=1,\ldots,t} \Phi_{i_b} \geq 0$. This will finish the proof of the lemma by what we said above. Let $M$ be the union of the indices occurring in $M_b$, $b = 1, \ldots, t$. We write

$$\Psi_k = -\sum\nolimits_{b=1,\ldots,t} \Psi_{i_b} + \Psi'_k$$

where

$$\begin{aligned} \Psi'_k = m_k w_k \Bigg( -1 + g_k + \frac{1}{2} \sum\nolimits_{b=1,\ldots t} \Big( \frac{a_{kj_{b,s_b}}}{w_k} - \frac{m_{i_b} w_{i_b}}{m_k w_k} \Big) + \frac{1}{2} \sum\nolimits_{l \neq k,\ l \notin M} \frac{a_{kl}}{w_k} \Bigg) \\ - \Bigg( -1 + \frac{1}{2} \sum\nolimits_{l \neq k,\ l \notin M,\ a_{kl} > 0} 1 \Bigg) \end{aligned}$$

Assume $\Psi'_k < 0$ to get a contradiction. If the set $\{l : l \neq k,\ l \notin M,\ a_{kl} > 0\}$ is empty, then $\{1, \ldots, n\} = M \cup \{k\}$ and $g_{top} = 0$ because $e = n - 1$ in this case and the result holds by Lemma 0.8. Thus we may assume there is at least one such $l$ which contributes $(1/2)a_{kl}/w_k \geq 1/2$ to the sum inside the first brackets. For each $b = 1, \ldots, t$ we have

$$\frac{a_{kj_{b,s_b}}}{w_k} - \frac{m_{i_b} w_{i_b}}{m_k w_k} = \frac{w_{i_b}}{w_k} (1 - \frac{m_{i_b}}{m_k})$$

This expression is $\geq \frac{1}{2}$ because $m_k \geq 2m_{i_b}$ by the previous paragraph and is $\geq 1$ if $w_k < w_{i_b}$. It follows that $\Psi'_k < 0$ implies $g_k = 0$. If $t \geq 2$ or $t = 1$ and $w_k < w_{i_1}$, then $\Psi'_k \geq 0$ (here we use the existence of an $l$ as shown above) which is a contradiction too. Thus $t = 1$ and $w_k = w_{i_1}$. If there at least two nonzero terms in the sum over $l$ or if there is one such $k$ and $a_{kl} > w_k$, then $\Psi'_k \geq 0$ as well. The final possibility is that $t = 1$ and there is one $l$ with $a_{kl} = w_k$. This is disallowed as this would mean $k \in Z$ contradicting the maximality of $M_1$. $\square$